AGV
AGRIVOLT PROTOCOL

Technical Audit & Infrastructure Verification Index v2025.10

(Security Audits, Oracle Verification, and Multi-Layer Architecture Overview)

I. Smart Contract Security Audit – Beosin Blockchain Lab (Aug 2025)

Document: AGV-NFT Smart Contract Security Audit.pdf **Audit No.: 202508271019**

No.	Audit Scope	Description / Result	Status
1	**NFT Minting Contracts (SeedPass / TreePass / SolarPass / ComputePass)**	ERC-721A upgradeable contracts with role-based AccessControl (Admin / Treasurer / Agent Minter).	Completed
2	**Findings**	1 Low risk + 2 Info, all fixed per report pages 8-10.	Resolved
3	**Audit Methodology**	Formal Verification + Manual Review + Static Analysis (VaaS Tool).	—
4	**Verified By**	Beosin Security Team, Aug 25–27 2025.	—

II. Main Token & Oracle Layer (Power-to-Mint Core)

1

Main Token Contract (GVT / rGGP)

ERC-20 core token controlling DAO Treasury, bonding curve, and reward logic.

Under internal review (Oct 2025)

2

Oracle Verification Layer

Syncs real-time solar generation data from Yichuan IoT → BVI SPV → Chain. Includes daily hash logs and SHA-256 signatures.

Operational (see "Daily Power Generation Snapshot")

3

Power Output Hash Integration

Monthly settlement records uploaded on-chain via Oracle to ensure proof-of-energy minting.

✅ Live since Oct 2025

III. Agent / Referral Distribution System (Layered Sales Architecture)

No.	Module	Description / Function
1	Multi-Tier Referral Engine	Three-level agent system linking on-chain NFT mint to off-chain partner dashboards. Calculates real-time commission ratios and distributes rGGP rewards. **Deployed Testnet v1.2**
2	Whitelist + Agent Integration	Agent-based minting function exempt from fees (SeedPass contract). **Audited (Beosin Finding [AGV-NFT-03])**
3	Treasurer Role Control	Multi-sig wallet manages royalty withdrawals and fee distribution. Implemented

IV. Multi-Language Access & Frontend Security

No.	Component	Description / Purpose	Status
1	**Internationalisation (i18n)**	Website & Investor Portal support EN / ZH / TH / VN interfaces for cross-border partners.	Implemented
2	**Frontend Security Audit**	Static scan for API keys, wallet connections (Metamask / OKX Wallet).	Internal Review Sep 2025
3	**Access Gateway Control**	Password-protected VC Data Room and NDA gate.	Live

V. Technical Roadmap & Integration Milestones

Phase	Timeline	Key Deliverables
Phase 1	Q3 2025	NFT Contracts + DAO Treasury Deployment (Beosin Audit Passed)
Phase 2	Q4 2025	Oracle Integration + Energy Data Hashing + Power-to-Mint Launch
Phase 3	Q1 2026	Layered Agent System + Multi-Chain Launch (BNB / Arbitrum / Polygon)
Phase 4	Q2 2026	Global Localisation + CEX Listing Compliance Audit

Smart Contracts

Foundation deployment

Power-to-Mint

Oracle integration

Multi-Chain

Cross-chain expansion

Global Launch

Compliance & listing

Verification Statement

All smart-contract and on-chain components have been reviewed under Beosin's security framework and AGV Protocol's internal audit team. Power-to-Mint data feeds are continuously verified by the Oracle layer and cross-checked against State Grid settlement records.

Security Assurance: This comprehensive audit framework ensures the integrity and reliability of the AGV Protocol infrastructure across all operational layers.